Emma L. Alexander

MPhys(Hons) · PhD · FRAS · AFHEA

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Summary

Astrophysicist with extensive experience delivering undergraduate physics and astronomy teaching, practical laboratory teaching, academic tutoring and student research supervision. Experienced in communicating complex physics concepts to students and public audiences through lectures, seminars, practicals, workshops and presentations. Research expertise in radio astronomy, polarisation, and machine learning, providing a strong research-led foundation for teaching across physics and astronomy. Associate Fellow of Advance HE (AFHEA).

Experience

University of Leeds — *Leeds, UK*

Research Fellow — *Nov 2024 - pres.*

- Data analysis and interpretation of non-thermal radio filaments in the Galactic plane (PI: Prof. Mark Thompson).
- Observation planning and scheduling with the MeerKAT telescope; reduction and imaging of JVLA data.

Government Statistical Service (GSS) — *Salford, UK*

Ad hoc Statistical Officer in HM Revenue & Customs (HMRC) — *Jan - Nov 2024*

- Provided statistical insights to improve HMRC data processes, using a variety of software including SQL and R.
- Aided in development of an evidence-based proposal to HM Treasury for investment in HMRC's Data and Analysis capabilities.
- Made a PowerBI dashboard to share with key stakeholders, highlighting analysis and key results.
- Personally recognised with internal award for collaboration.

University of Manchester — *Manchester, UK*

Postdoctoral Research Associate — *Oct 2021 - Dec 2023*

- Developed machine-learning workflows in Python for astronomical classification problems, including dataset construction, preprocessing, feature engineering and collaboration with industrial AI partners.
- Co-builder of Zooniverse citizen science project Radio Galaxy Zoo: EMU.

PhD student (Supervisors: Dr. Patrick Leahy & Prof. Anna Scaife) — *Sept 2017 - Sept 2021*

- Thesis title: Magnetic Fields around Radio Galaxies. Analysis of large polarisation data cubes from the Australian Square Kilometre Array Pathfinder telescope (ASKAP) via extensive use of programming languages (e.g. Python) and astronomical software (e.g. CASA) in both Linux and MacOSX environments. Observing with the Australia Telescope Compact Array (ATCA): proposal writing (as PI & Co-I), observation planning and supervising, reduction and analysis of data in Miriad.

Undergraduate research — *Various, 2015 - 2017*

- Calibration and imaging of C, L, and X-band JVLA data in CASA of NGC 628; analysis of Faraday rotation maps with Python scripts.
- Reduction and analysis of ATCA data of the magnetically chemically peculiar star CU Virginis in Miriad; imaging of radio pulses from the star in CASA, and spectrographic analysis of the results via Python.
- Solar sidelobe analysis for the C-Band All Sky Survey (C-BASS); improved existing models of sidelobes and solar variability using IDL, Python and MATLAB.

Netherlands Institute for Radio Astronomy (ASTRON) — *Dwingeloo, Netherlands*

Summer Student (Supervisor: Dr. Jess Broderick) — *June - Sept 2017*

- Normalisation and stacking of uncalibrated data from an exploratory scan by the Engineering Development Array (EDA), to obtain detections of carbon and hydrogen Radio Recombination Lines (RRLs) towards the Galactic Centre.

Education

University of Manchester — *Manchester, UK*

PhD Astronomy & Astrophysics — *2022*

MPhys Physics with Astrophysics (first class honours) — *2017*

Awards, Funding, and Professional Memberships

2025 **UKRI Public Engagement Spark Award (£20,000)**, Co-lead on Moonsighter's Academy ("MoonBack")
2023 **Winton Poster Prize (Postdoc)**, National Astronomy Meeting, Cardiff
2020 **Associate Fellowship of Advance HE (AFHEA)**
2017 **George Rigg Studentship**, Jodrell Bank Centre for Astrophysics
2017 **President's Doctoral Scholarship**, University of Manchester
2016 **Fellowship of the Royal Astronomical Society (FRAS)**

Teaching & Supervision

University of Leeds *Leeds, UK*

DEMONSTRATOR, PHYS3002 ADVANCED TECHNIQUES IN ASTROPHYSICS (RADIO INTERFEROMETRY) *2025 – present*

- Deliver practical teaching in radio interferometry to undergraduate astrophysics students, supporting students in applying specialist astronomical software and data-analysis techniques.
- Explain technical concepts and support students in developing independent approaches to practical data-analysis problems.

University of Manchester *Manchester, UK*

SUPERVISOR, MPHYS STUDENT RESEARCH PROJECT *2023*

- Supervised an MPhys research project, "Predicting the visibility of the New Crescent Moon with Machine Learning".
- Supported the student in developing research questions, applying computational methods and interpreting scientific results.

University of Manchester *Manchester, UK*

ACADEMIC TUTOR, 2ND YEAR PHYSICS *2022 – 2023*

- Supported five second-year physics undergraduates across the academic year, planning and delivering weekly tutorials to reinforce understanding of core modules.
- Used continuous formative assessment to identify learning needs and contribute to students' overall grades.
- Provided targeted one-to-one support, facilitated peer learning, identified common misunderstandings and developed additional resources.

University of Manchester *Manchester, UK*

GRADUATE TEACHING ASSISTANT, UNDERGRADUATE PHYSICS LABORATORIES *2017 – 2021*

- Delivered laboratory teaching to first- and second-year physics undergraduates, introducing experiments and supporting students with background theory, software and practical techniques.
- Worked with approximately 12 students at a time across 3–4 week experiments, culminating in interview-style final assessment.
- Marked undergraduate laboratory reports and provided written feedback through Blackboard.
- Supported students with differing needs, including accessibility and pastoral requirements.
- GTA representative on the Undergraduate Laboratory Teaching Committee, contributing to the transition to online teaching during COVID-19.
- Completed GTA training in teaching skills, learning outcomes and lesson planning, e-learning and Blackboard, and marking and feedback.
- Professional development in evaluation and impact, reaching underserved audiences, schools outreach, and sharing educational practice.

University of Manchester *Manchester, UK*

PASS LEADER & PEER MENTOR *2014 – 2016*

- Led weekly peer-learning sessions for first-year physics students, providing support with physics problems, study skills and transition to university.
- Facilitated group projects, supporting students with collaboration, workload management and delegation.

Presentations

Non-thermal Radio Filaments in the Galactic Plane

- National Astronomy Meeting (contributed talk, Durham, 2025)
- *A new era in astrophysics: Preparing for early science with the SKAO (poster & sparkler, Görlitz, 2025)*

Talks about PhD research: Magnetic fields around radio galaxies with POSSUM

- 2023: *National Astronomy Meeting (poster & sparkler, Cardiff)*; *'New Eyes on the Universe' conference (poster, remote)*
- 2022: SKA Magnetism SWG meeting (**invited talk**, remote); SPARCS XI conference (contributed talk, remote)
- 2021: Curtin University colloquium (**invited talk**, remote); SPARCS X conference (contributed talk, remote); 'A precursor view of the SKA Sky' conference (contributed talk, remote); Jodrell Bank Centre for Astrophysics (internal seminar)
- 2019: *'New Science enabled by New Technologies in the SKA Era (poster)*

Overview talks: Radio Astronomy & Astrophysical Magnetism

- North American Foundation Awards for Postgraduate Study at the University of Manchester (NAFUM) board meeting (**invited**, remote, 2020).
- PhD discussion sessions and work experience weeks (various contributions, 2018-2020)

Other academic talks

- Moonsighters Academy: Connecting the British Muslim Community to the Cosmos, National Astronomy Meeting (Birmingham, 2026)
- The New Crescent Moon and its role in the Islamic calendar, National Astronomy Meeting (Cardiff, 2023)
- Recombination line science with SKA precursor technology (Astrolunch, ASTRON, Netherlands, 2017)

Publications

Refereed

13. S. N. Bradbury et al. (inc. **E. L. Alexander**), **Helical radio jets as probes of magnetised cluster environments: Periodic Faraday Rotation Revealed in the Corkscrew Galaxy by POSSUM**, MNRAS (2026).
12. A. M. Hopkins et al. (inc. **E. L. Alexander**), **The Evolutionary Map of the Universe: A new radio atlas for the southern hemisphere sky**, PASA 42 e071 (2025).
11. B. M. Gaensler et al. (inc. **E. L. Alexander**), **The Polarisation Sky Survey of the Universe's Magnetism (POSSUM). I. Science Goals and Survey Description** PASA (2025).
10. L. Rudnick et al. (inc. **E. L. Alexander**), **Pseudo-3D visualization of Faraday structure in polarized radio sources: methods, science use cases, and development priorities** MNRAS 535, 3, 2115–2128 (2024).
9. I. McDonald et al. (inc. **E. L. Alexander**), **PySSED: an automated method of collating and fitting stellar spectral energy distributions.** RAS Techniques and Instruments (2024): rzae005.
8. S. E. Cody et al. (inc. **E. L. Alexander**), **Machine learning based stellar classification with highly sparse photometry data.**. Open Research Europe 4 (2024): 29.
7. M. M. Boyce et al. (inc. **E. L. Alexander**), **Hydra II: Characterisation of Aegean, Caesar, ProFound, PyBDSF, and Selavy source finders.** PASA 1-28 (2023).
6. M. M. Boyce et al. (inc. **E. L. Alexander**), **Hydra I: An extensible multi-source-finder comparison and cataloguing tool.** PASA 1-27 (2023).
5. M. Bowles, H. Tang, E. Vardoulaki, **E. L. Alexander** et al., **Radio Galaxy Zoo EMU: Towards a Semantic Radio Galaxy Morphology Taxonomy**. MNRAS 522, 2, 2584–2600 (2023).
4. G. Segal et al. (inc. **E. L. Alexander**, **Identifying anomalous sources in the EMU Pilot Survey data using a complexity-based approach.**. MNRAS 521, 1, 1429-1447 (2023).
3. M. Cárcamo, A. M. M. Scaife, **E. L. Alexander** and J. P. Leahy, **CS-ROMER: A novel compressed sensing framework for Faraday depth reconstruction.** MNRAS 518, 2, 1955-1974. (2023).
2. T. D. Joseph, M. D. Filipović, E. J. Crawford, I. Bojičić, **E. L. Alexander** et al., **The ASKAP-EMU Early Science Project: Radio Continuum Survey of the Small Magellanic Cloud**. MNRAS 490, 1, 1202-1219 (2019).
1. J. B. R. Oonk, **E. L. Alexander**, J. Broderick, M. Sokolowski, and R. Wayth, **Spectroscopy with the Engineering Development Array: cold H+ at 63 MHz towards the Galactic center**. MNRAS 487, 4, 4737–4750 (2019).

Conference papers

2. E. Vardoulaki et al. (inc. **E. L. Alexander**), **Radio Galaxy Zoo EMU: Harnessing Citizen Science and AI to Advance Open Science Catalogues**. Proceedings IAU397 Symposium UniversAI Exploring the Universe with Artificial Intelligence.
1. M. Bowles, H. Tang, E. Vardoulaki, **E. L. Alexander** et al., **A New Task: Deriving Semantic Class Targets for the Physical Sciences**. Fifth Workshop on Machine Learning and the Physical Sciences (NeurIPS 2022). `https://arxiv.org/abs/2210.14760`

Service

UK SKAO Early Career Researchers Committee, Member	2026 – pres.
Institute of Physics Careers Panel, Dept. of Physics & Astronomy, University of Manchester, UK	2024
Laboratory teaching committee, Dept. of Physics & Astronomy, University of Manchester, UK	2020 – 2021
Postgraduate representative, Dept. of Physics & Astronomy, University of Manchester, UK	2018 – 2020
PhD interviews support team, Jodrell Bank Centre for Astrophysics, University of Manchester	2020
Local Organising Committee: Internal Symposium, Jodrell Bank Centre for Astrophysics, University of Manchester	2019
Local Organising Committee: A Centenary of Astrophysical Jets conference, SKAO HQ, Cheshire, UK	2019
Laboratory open day tour guide, University of Manchester, UK	2019
Internal Seminar organiser, Jodrell Bank Centre for Astrophysics, University of Manchester, UK	2017 – 2018
Postgraduate committee, Jodrell Bank Centre for Astrophysics, University of Manchester, UK	2017 – 2018
Astronomy Society Committee, (Science Officer; then Chair; then Secretary), University of Manchester, UK	2014 – 2017
Physics Netball team, Captain , University of Manchester, UK	2015 – 2016
UCAS interview day help, School of Physics & Astronomy, University of Manchester, UK	2015 – 2016

Selected Media & Outreach

Talks, e.g. visiting local astronomical societies to give 'A Tour of the Radio Universe' and other talks.
Television, e.g. multiple live appearances (in-studio and remote) to discuss astronomy news on BBC Breakfast.
Radio, discussion of astronomical topics on national & local radio.
Podcasting, previously a regular contributor to the *Jodcast* astronomy podcast.
Cruises, Go Stargazing astronomer on selected Fred Olsen cruises.
Events, e.g. Bluedot festival and ScienceX.
Schools, e.g. workshops with IntoUniversity.
Articles, e.g. "A 4G network on the Moon is bad news for radio astronomy", The Conversation, 2020.
Consulting, e.g. astronomy content for *Netflix: Night on Earth* (episode: *Moonlit Plains*).